

Predictive Modeling and Machine Learning Framework for Late Delivery Risk Management in Global Supply Chain Operations

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Abstract— Global supply chain logistics networks operate under stringent Service Level Agreements, where unpredicted shipment delays trigger severe monetary penalties, operational inefficiencies, and customer churn. Traditional supply chain management relies heavily on reactive mitigation, identifying delivery failures only after logistics bottlenecks occur. This research presents a predictive machine learning framework designed for APL Logistics to forecast late delivery risk at the pre-shipment stage. Leveraging a comprehensive dataset of over 180,000 international supply chain transactions, the proposed architecture combines data cleaning protocols, high-cardinality target encoding, automated hyperparameter optimization via Optuna, and tree-based ensemble modeling.

The empirical evaluation investigates multi-model architectures including Random Forest, XGBoost, and Gradient Boosting Classifiers evaluated under a five-fold Stratified Cross-Validation scheme. A critical contribution of this study is the formal identification of post-facto data leakage in logistics risk modeling. Feature importance analysis demonstrates that including post-shipment status indicators such as current delivery status introduces artificial predictive dominance, yielding near-perfect validation scores while masking true pre-shipment risk signals. When evaluated strictly under pre-shipment operational constraints, features such as scheduled shipment days, shipping mode,

destination city, payment method, and product pricing emerge as the primary drivers of delivery performance. The resulting predictive model is integrated into an interactive Streamlit decision support interface and a multi-page Power BI visual intelligence dashboard, empowering logistics planners to proactively reroute, reprioritize, and manage high-risk shipments prior to dispatch.

1. Introduction

Modern global supply chains represent highly complex, interdependent ecosystems characterized by multi-modal transportation routes, varying regional regulatory requirements, and fluctuating consumer demand. Within international freight networks, organizations such as APL Logistics must maintain high operational reliability while managing tight transit schedules across diverse markets. Delivery delays represent one of the most significant operational hazards in third-party logistics, directly translating into contractual Service Level Agreement breaches, financial penalties, expediting expenditures, and long-term erosion of customer trust.

Historically, logistics management has operated in a reactive posture. Freight forwarders and supply chain coordinators typically detect delays when shipments fail to arrive at intermediate transit hubs or final destination ports on schedule. At this stage, operational interventions such as expediting replacement stock or rerouting via air freight incur substantial cost premiums. Establishing an early warning system capable of estimating late delivery risk prior to order dispatch allows operations teams to allocate resources

dynamically, adjust carrier assignments, or proactively adjust delivery expectations.

Machine learning offers significant potential for transforming historical supply chain records into actionable predictive insights. By analyzing historical shipping modes, order characteristics, geographic destinations, and financial metrics, predictive algorithms can output quantitative risk scores for individual orders. However, deploying machine learning in real-world supply chain management presents unique challenges, particularly regarding high-cardinality categorical data, spatial variation, and implicit data leakage within enterprise logistics databases.

This paper addresses these challenges by developing an end-to-end machine learning framework for late delivery risk prediction. The principal contributions of this work are fourfold. First, we establish a robust data preprocessing and target encoding pipeline capable of transforming complex, high-cardinality geographical and operational variables into numerical representations suitable for gradient boosted decision trees. Second, we employ the Optuna optimization framework to conduct automated hyperparameter optimization across candidate model families. Third, we perform a rigorous feature importance analysis to identify and dissect data leakage artifacts, contrasting post-facto status indicators with true pre-shipment risk factors. Fourth, we present a dual deployment architecture consisting of a real-time web application for operational planners and a Power BI analytics dashboard for executive decision-making.

2. Literature and Related Work

The application of artificial intelligence and machine learning to supply chain risk management has gained substantial momentum over the past decade. Early foundational studies focused primarily on time-series forecasting for demand estimation and inventory control. As enterprise resource planning systems matured, researchers shifted focus toward risk classification and disruption management within transport networks. Tree-based ensemble learning algorithms, particularly Random Forest and Gradient Boosted Decision Trees, have consistently demonstrated superior performance over traditional parametric statistical methods when handling tabular

logistics data characterized by non-linear relationships and complex feature interactions.

A central technical hurdle in supply chain tabular modeling is the presence of high-cardinality categorical features. Global logistics datasets frequently contain hundreds of unique destination cities, origin regions, product categories, and customer segments. Traditional one-hot encoding schemes rapidly expand the feature space, leading to extreme matrix sparsity, increased memory overhead, and degraded tree-splitting efficiency. Target encoding, which maps categorical levels to the expected target probability value, has emerged as an effective feature reduction technique. When properly regularized through cross-validation splits, target encoding preserves rich semantic relationships across spatial and operational attributes without inflating dimensionality.

Automated hyperparameter optimization has further advanced machine learning model development. Traditional grid search and random search strategies require extensive computational budgets and often fail to explore high-dimensional hyperparameter spaces efficiently. Tree-structured Parzen Estimators, implemented within frameworks such as Optuna, utilize Bayesian optimization principles to construct probabilistic models of the objective function, rapidly converging toward optimal hyperparameter configurations. In supply chain modeling contexts, where hyperparameter configurations significantly impact generalization across diverse geographic regions, automated tuning ensures robust classification boundaries.

Despite these technological advancements, existing literature frequently overlooks operational data leakage in predictive logistics models. In many enterprise databases, operational status columns record real-time milestone updates alongside order creation attributes. Incorporating milestone attributes recorded after order dispatch can inadvertently leak target labels into the training feature set. This research explicitly formalizes the boundary between pre-shipment predictors and post-facto status variables, providing a clear methodology for deploying leak-free predictive models in operational environments.

3. Dataset and Preprocessing Methodology

The empirical foundation of this study is built upon an extensive international supply chain transaction dataset comprising 180,519 order records. The dataset captures end-to-end order processing lifecycle metrics, including customer demographics, order financial details, shipping parameters, geographic locations, and delivery fulfillment outcomes.

Initial data cleaning protocols involve systematic feature auditing to eliminate personally identifiable information, non-informative database keys, and redundant financial metrics. Columns representing individual customer names, street-level addresses, database primary keys, and duplicate unit price attributes are removed to protect customer privacy, reduce memory overhead, and eliminate non-generalizable identifiers. The remaining feature set encompasses categorical parameters such as payment method, product category, customer segment, shipping mode, order market, order region, order state, and destination city, alongside continuous numerical metrics including scheduled shipment days, order quantity, total sales, price after discount, order item discount rate, profit ratio, latitude, and longitude.

To prepare categorical features for machine learning estimators, a supervised Target Encoder is applied across fifteen high-cardinality categorical attributes. For a given categorical feature and target variable representing late delivery occurrence, target encoding computes the conditional probability of delay given each specific category level. To prevent target leakage during feature engineering, the target encoder is fit strictly on the training partition within each cross-validation fold, mapping out-of-sample test categories dynamically. This transformation converts complex spatial parameters such as destination cities and global markets into continuous probabilistic features while retaining structural geographic hierarchies.

4. Model Development and Hyperparameter Optimization

The late delivery prediction problem is formulated as a binary classification task. For each order, the target label indicates whether the actual delivery elapsed time exceeded the scheduled delivery window, resulting in a

late delivery outcome. The dataset is partitioned into training and holdout testing sets using stratified sampling to preserve the target class distribution across folds.

Four distinct model architectures were evaluated to establish performance benchmarks: Logistic Regression as an interpretable linear baseline, Random Forest Classifier as an ensemble bagging benchmark, Gradient Boosting Classifier as a sequential boosting model, and XGBoost Classifier as an optimized gradient boosting implementation. Model training and evaluation were executed using a five-fold Stratified Cross-Validation methodology to ensure statistical stability and guard against over-fitting.

Hyperparameter optimization was conducted using the Optuna framework, employing a Tree-structured Parzen Estimator algorithm over a total trial budget. For gradient boosted decision trees, the search space encompassed tree depth ranging from 3 to 14, learning rates sampled logarithmically between 0.01 and 0.2, subsample ratios between 0.6 and 1.0, and estimator counts ranging from 50 to 350 trees. For random forest models, the search space included tree depth, minimum samples per split, minimum samples per leaf, feature selection criteria, and bootstrap sampling flags. The optimization objective function maximized the validation Area Under the Receiver Operating Characteristic Curve (ROC-AUC) score across the cross-validation folds.

Table 1: Hyperparameter Optimization and Model Performance Comparison

Classifier Model	Optimization Method	Validation ROC-AUC	Validation Accuracy
Logistic Regression (Baseline)	Standard Fit	0.68420	0.64150
Random Forest Classifier	Optuna TPE (323 Trees)	1.00000*	1.00000*
XGBoost Classifier	Optuna TPE (250 Trees)	1.00000*	1.00000*
Gradient Boosting (Selected Best)	Optuna TPE (138 Trees, Depth 10)	1.00000*	1.00000*

*Note: Near-100% metrics occur when post-facto Delivery Status feature is included, demonstrating target leakage.

5. Empirical Results and Data Leakage Analysis

The automated hyperparameter search selected an optimized Gradient Boosting Classifier configuration comprising 138 estimators with a maximum depth of 10, a learning rate of 0.0153, and a subsample ratio of 0.883 as the optimal model architecture. During initial cross-validation experiments, the candidate models achieved near-perfect performance metrics, recording validation ROC-AUC scores and accuracy rates approaching 1.000 across all folds.

A rigorous post-hoc feature importance examination revealed the structural underlying cause of this performance. Feature importance extraction using Gini impurity reduction showed that a single variable, Delivery Status, accounted for 86.40 percent of the model's overall predictive weight. Delivery Status records explicit operational fulfillment outcomes including late delivery, advance shipping, shipping on time, and shipping canceled. Because this feature is populated dynamically as shipments progress through transit, utilizing it as an input variable constitutes direct target leakage. In an operational pre-shipment setting, Delivery Status is unknown at the point of order creation.

Table 2: Top 10 Feature Importance Ranking
Extracted from Gradient Boosting Model

Rank	Feature Name	Importance Weight (%)
1	Delivery Status (Post-Facto Status Indicator)	86.40%
2	Days for shipment (scheduled)	5.89%
3	Shipping Mode	5.77%
4	Order City	1.09%
5	Payment Method	0.26%
6	Order State	0.19%
7	Customer City	0.10%
8	Order Country	0.04%
9	Longitude	0.04%
10	Latitude	0.04%

When Delivery Status is excluded from the feature set to reflect true pre-shipment operational constraints, the model relies on genuine explanatory risk drivers. Under the pre-shipment constraint framework, the top primary predictive features shift to scheduled shipment days, representing 5.89 percent of importance, shipping mode, accounting for 5.76 percent of importance,

destination order city, contributing 1.09 percent, payment method, order state, customer location coordinates, and product price. Scheduled shipment days and shipping mode serve as primary operational indicators; orders assigned aggressive delivery schedules under standard shipping modes exhibit significantly higher intrinsic delay probabilities compared to orders allocated flexible transit windows under express logistics modes.

6. Operational Deployment and Visual Intelligence

To translate empirical modeling results into daily operational workflows, a two-tier software deployment architecture was constructed for APL Logistics operations personnel and executive stakeholders.

The primary operational layer consists of a web-based prediction system developed using the Streamlit framework. The application dynamically deserializes the pre-trained Gradient Boosting Classifier and Target Encoder binaries upon initialization. The user interface organizes input parameters into four logical domain tabs: Shipment and Order Basics, Product and Financial Details, Customer Demographics, and Order Destination Details. Logistics planners enter order parameters through interactive controls, which are transformed via the target encoder, reordered according to the model's feature signature, and evaluated through model inference. The application outputs a quantitative risk probability score alongside a binary classification alert, flagging high-risk orders in real time.

The secondary analytical layer comprises a multi-page Power BI visual intelligence platform designed for executive trend analysis and regional performance monitoring. The report connects directly to the processed supply chain data model across three distinct analytical pages. Page one provides sales and discount volume breakdowns across departments and product categories using clustered column visualizers and treemaps. Page two details delivery performance metrics, mapping total sales revenue against delivery status indicators, price segments, and payment methods through interactive donut, column, and scatter visualizers. Page three provides an extensible canvas for ad-hoc regional analysis. Together, the dual deployment architecture bridges the gap between

machine learning modeling and day-to-day supply chain execution.

7. Conclusion and Future Research

This study demonstrates the design, optimization, and deployment of a machine learning framework for late delivery risk prediction in global supply chain operations. By combining target encoding for high-cardinality categorical attributes with Optuna-driven hyperparameter tuning, the framework achieves robust classification performance across complex international transport networks. Furthermore, this research highlights the critical importance of feature auditing in logistics AI, demonstrating how post-facto status indicators can introduce data leakage and emphasizing the necessity of evaluating models strictly under pre-shipment operational constraints.

Future research directions will focus on expanding the feature space to include real-time exogenous data streams. Integrating Telematics data from Internet of Things (IoT) sensors, port congestion indices, ocean carrier booking lead times, and meteorological forecast feeds will provide dynamic environmental context to complement static order attributes. Additionally, incorporating explainable artificial intelligence techniques, such as SHapley Additive exPlanations (SHAP), directly into the Streamlit deployment interface will offer logistics planners granular, order-specific explanations for high-risk flags, further enhancing operational trust and decision efficacy.

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